1-rdt-initial-data-exploration

February 23, 2024

1 1-rdt-initial-data-exploration

[ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[ ]:
```
# set seaborn style
sns.set_style("white")
sns.set_context("talk")
```

[ ]:
```
df = pd.read_csv("../data/processed/corpus_studio_processed.csv", sep=",")
df.head()
```

[ ]:
```
       word previous_word next_word      artist song_type  \
0  hollered     conductor       NaN  cannedheat     cover
1     found           NaN       NaN  jeffhealey     cover
2        my           NaN       old  jeffhealey     cover
3        my           NaN       NaN  jeffhealey     cover
4     there           NaN        is  albertking     cover

                         song            aae_feature  aae_realization  \
0            bring_it_on_home     post-consonantal d                0
1             blue_jean_blues     post-consonantal d                1
2             blue_jean_blues  ai monophthongization                1
3             blue_jean_blues  ai monophthongization                1
4  as_the_years_go_passing_by         post-vocalic r                0

  performance_type   time social_group             group
0           studio  1960s     nonAA_US     1960snonAA_US
1           studio  1980s  nonAA_nonUS  1980snonAA_nonUS
2           studio  1980s  nonAA_nonUS  1980snonAA_nonUS
3           studio  1980s  nonAA_nonUS  1980snonAA_nonUS
4           studio  1960s           AA           1960sAA
```

[ ]:
```
pd.crosstab(index=df["aae_feature"], columns=df["aae_realization"]).sort_values(
    by=1, ascending=False
```

```
)
```

[ ]:
```
aae_realization           0     1
aae_feature
ai monophthongization  1157  4731
post-vocalic r          870  2177
post-consonantal d      614  1404
ing ultimas             147  1217
post-consonantal t      795  1191
auxiliary verb            3   250
third person singular   138    92
zero copula             258    55
```

[ ]:
```
pd.crosstab(index=df["artist"], columns=df["aae_realization"]).sort_values(
    by=1, ascending=False
)
```

[ ]:
```
aae_realization         0    1
artist
albertcollins          92  367
thebluesband           92  346
kokotaylor             93  333
davyknowles            99  327
shemekiacopeland      110  321
robbenford            137  306
philipsayce            86  304
rorygallagher          83  298
tinylegstim            85  294
savoybrown            116  275
jeffhealey            106  273
muddywaters            58  272
joebonamassa           92  270
lutherallison          87  270
samanthafish           87  266
johnmayall             74  265
fabulousthunderbirds   89  257
allmanbrothers         73  254
jimihendrix            66  252
stevierayvaughan       51  251
johnleehooker          49  250
mattschofield          84  249
freddieking           119  248
garyclarkjr            48  248
joanneshawtaylor      112  241
janisjoplin           198  241
bonnieraitt            83  234
albertking             57  228
```

```
kingfish                 59  224
paulbutterfield         110  223
allyvenable             103  218
jjcale                   58  214
robertcray              119  214
cream                   105  213
tenyearsafter            68  211
daniwilde                68  203
ericgales                31  203
fleetwoodmac             66  199
kirkfletcher            122  187
danpatlansky             48  187
cannedheat              119  184
rollingstones            53  177
bbking                   98  177
ericclapton             151  173
stevemiller              78  170
```

[ ]:
```
pd.crosstab(
    index=df["artist"], columns=df["aae_realization"], normalize="index"
).sort_values(by=1, ascending=False)
```

[ ]:
```
aae_realization              0         1
artist
ericgales             0.132479  0.867521
garyclarkjr           0.162162  0.837838
johnleehooker         0.163880  0.836120
stevierayvaughan      0.168874  0.831126
muddywaters           0.175758  0.824242
albertking            0.200000  0.800000
albertcollins         0.200436  0.799564
danpatlansky          0.204255  0.795745
jimihendrix           0.207547  0.792453
kingfish              0.208481  0.791519
thebluesband          0.210046  0.789954
jjcale                0.213235  0.786765
rorygallagher         0.217848  0.782152
johnmayall            0.218289  0.781711
kokotaylor            0.218310  0.781690
philipsayce           0.220513  0.779487
allmanbrothers        0.223242  0.776758
tinylegstim           0.224274  0.775726
rollingstones         0.230435  0.769565
davyknowles           0.232394  0.767606
lutherallison         0.243697  0.756303
tenyearsafter         0.243728  0.756272
samanthafish          0.246459  0.753541
```

```
fleetwoodmac          0.249057  0.750943
daniwilde             0.250923  0.749077
mattschofield         0.252252  0.747748
joebonamassa          0.254144  0.745856
shemekiacopeland      0.255220  0.744780
fabulousthunderbirds  0.257225  0.742775
bonnieraitt           0.261830  0.738170
jeffhealey            0.279683  0.720317
savoybrown            0.296675  0.703325
robbenford            0.309255  0.690745
stevemiller           0.314516  0.685484
joanneshawtaylor      0.317280  0.682720
allyvenable           0.320872  0.679128
freddieking           0.324251  0.675749
cream                 0.330189  0.669811
paulbutterfield       0.330330  0.669670
bbking                0.356364  0.643636
robertcray            0.357357  0.642643
cannedheat            0.392739  0.607261
kirkfletcher          0.394822  0.605178
janisjoplin           0.451025  0.548975
ericclapton           0.466049  0.533951
```

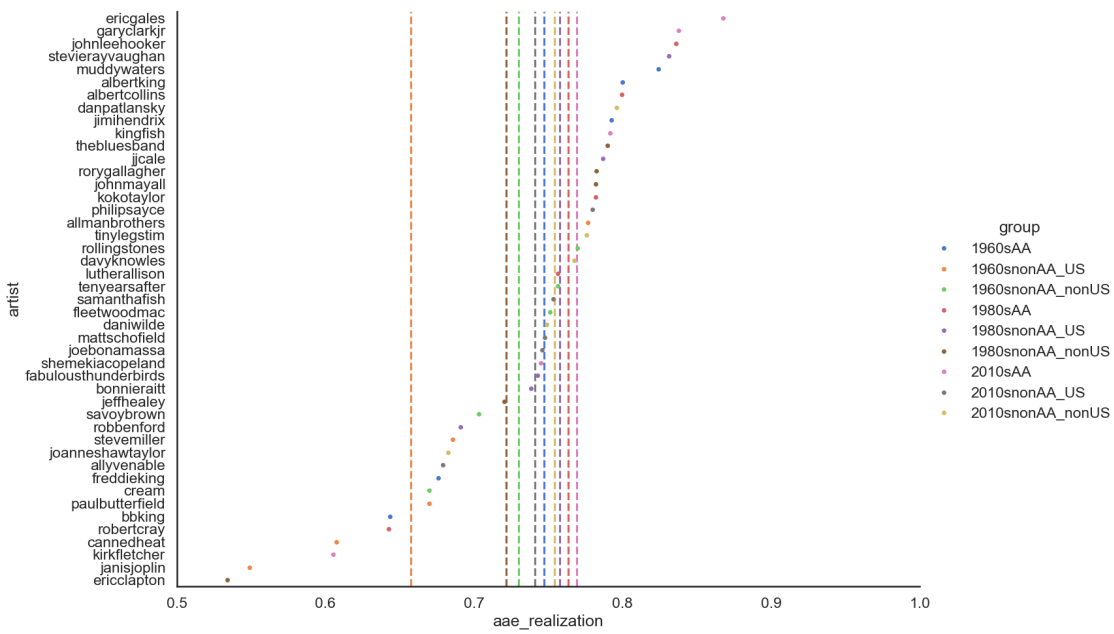
[ ]:
```python
# calculate group means
blues_artist_mean = (
    df.groupby(["artist", "group"])["aae_realization"]
    .mean()
    .reset_index()
    .sort_values(by="aae_realization", ascending=False)
)

# define custom colors for each group
custom_colors = sns.color_palette("muted", 9)

# create the catplot
g = sns.catplot(
    data=blues_artist_mean,
    x="aae_realization",
    y="artist",
    hue="group",
    hue_order=[
        "1960sAA",
        "1960snonAA_US",
        "1960snonAA_nonUS",
        "1980sAA",
        "1980snonAA_US",
        "1980snonAA_nonUS",
```

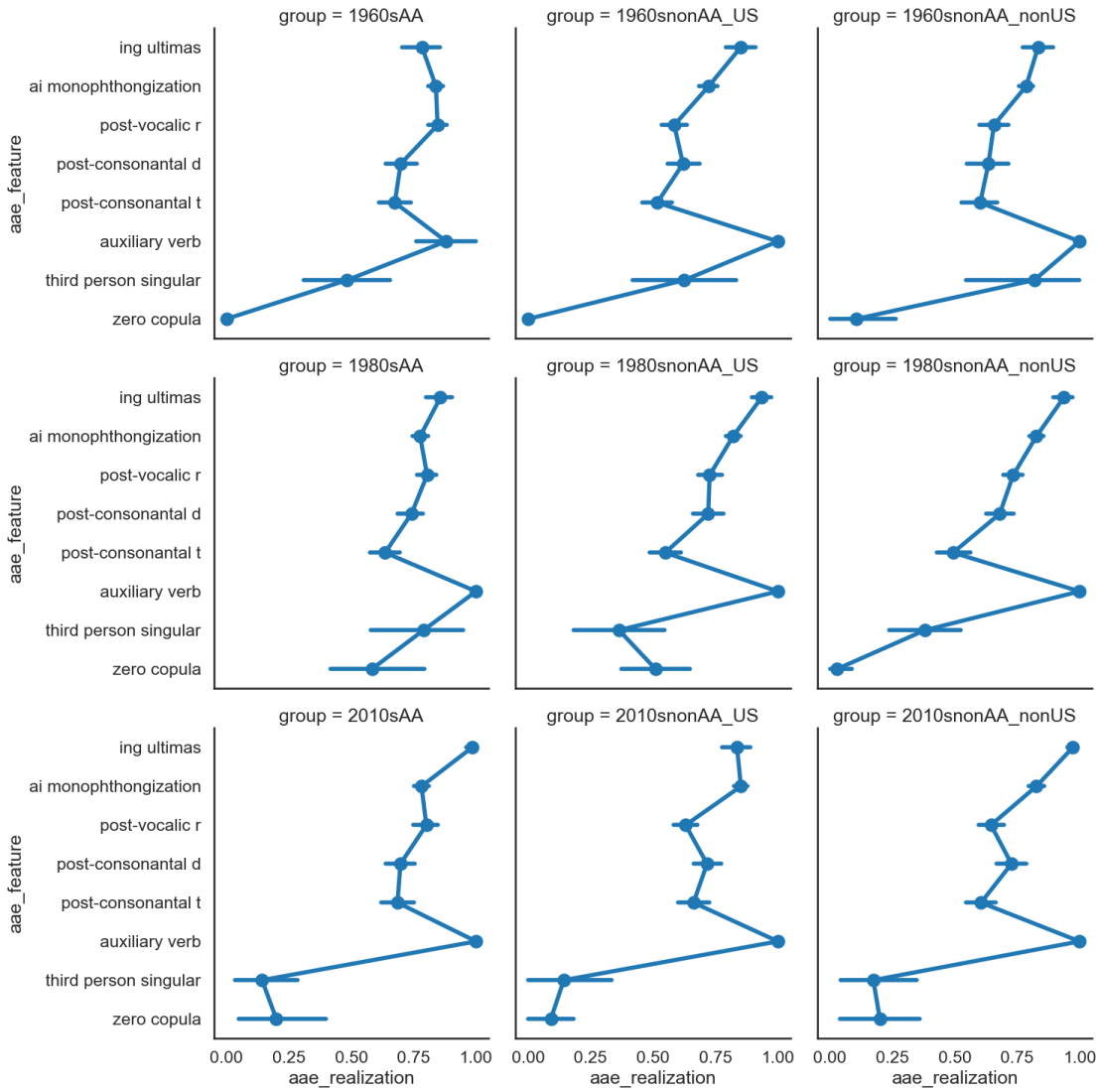
```
        "2010sAA",
        "2010snonAA_US",
        "2010snonAA_nonUS",
    ],
    palette=custom_colors,
    height=10,
    aspect=1.5,
)
# calculate group means and add as horizontal lines
group_means = blues_artist_mean.groupby("group")["aae_realization"].mean()
for group, color in zip(group_means.index, custom_colors):
    mean_value = group_means[group]
    plt.axvline(mean_value, color=color, linestyle="--")

# show the plot
plt.xlim(0.5, 1)
plt.savefig("../reports/figures/mean_aae_realizations.png", dpi=600)
plt.show()
```



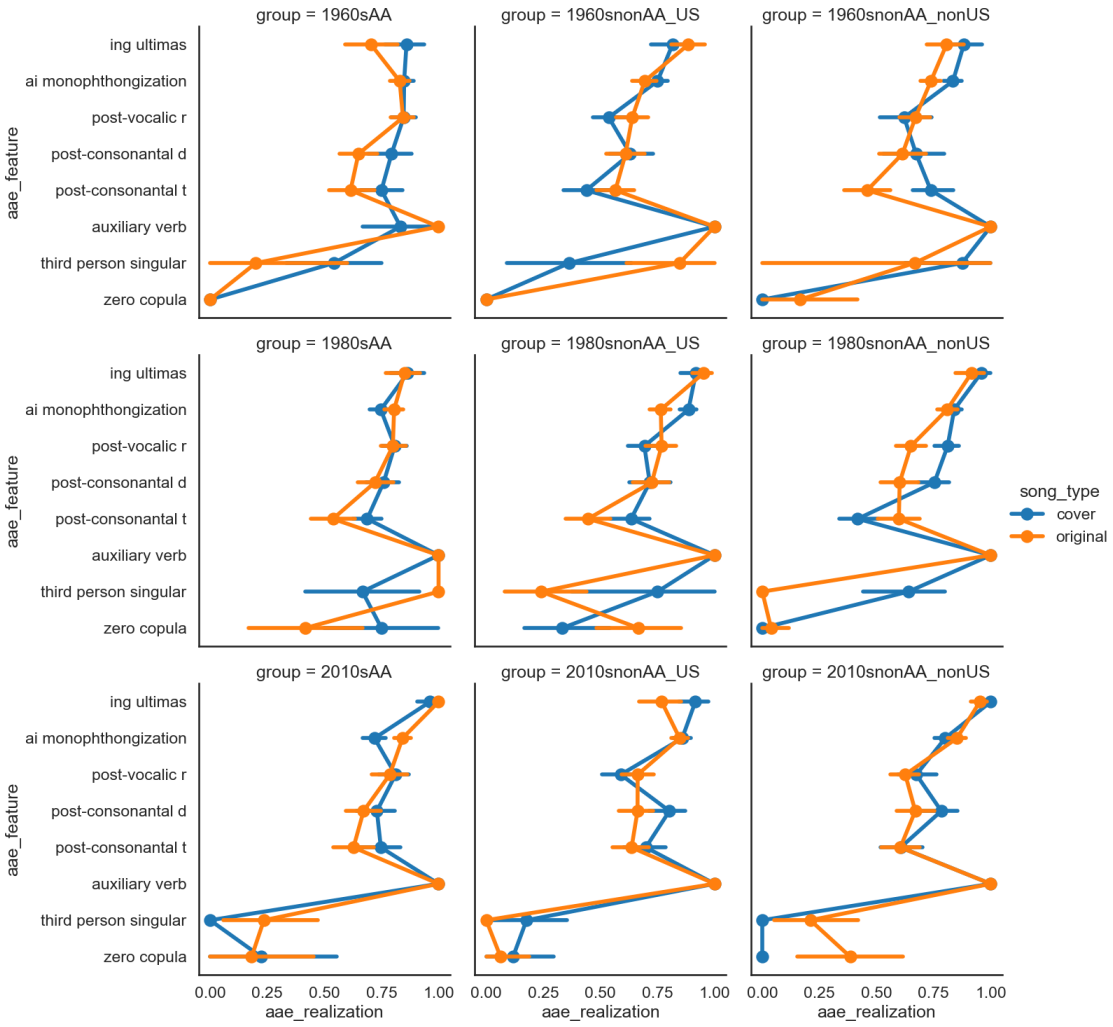
[ ]:
```
sns.catplot(
    data=df,
    y="aae_feature",
    x="aae_realization",
    col="group",
    col_wrap=3,
    col_order=[
```

```
        "1960sAA",
        "1960snonAA_US",
        "1960snonAA_nonUS",
        "1980sAA",
        "1980snonAA_US",
        "1980snonAA_nonUS",
        "2010sAA",
        "2010snonAA_US",
        "2010snonAA_nonUS",
    ],
    order=[
        "ing ultimas",
        "ai monophthongization",
        "post-vocalic r",
        "post-consonantal d",
        "post-consonantal t",
        "auxiliary verb",
        "third person singular",
        "zero copula",
    ],
    kind="point",
    errorbar="ci",
)
plt.savefig("../reports/figures/point_plots_by_group.png", dpi=600)
plt.show()
```



[ ]:
```
sns.catplot(
    data=df,
    y="aae_feature",
    x="aae_realization",
    col="group",
    col_wrap=3,
    hue="song_type",
    col_order=[
        "1960sAA",
        "1960snonAA_US",
        "1960snonAA_nonUS",
        "1980sAA",
        "1980snonAA_US",
```

```
        "1980snonAA_nonUS",
        "2010sAA",
        "2010snonAA_US",
        "2010snonAA_nonUS",
    ],
    order=[
        "ing ultimas",
        "ai monophthongization",
        "post-vocalic r",
        "post-consonantal d",
        "post-consonantal t",
        "auxiliary verb",
        "third person singular",
        "zero copula",
    ],
    kind="point",
    errorbar="ci",
)
plt.savefig("../reports/figures/point_plots_by_group_and_song_type.png",␣
 ↪dpi=600)
plt.show()
```


group = 1960sAA
group = 1960snonAA_US
group = 1960snonAA_nonUS
group = 1980sAA
group = 1980snonAA_US
group = 1980snonAA_nonUS
group = 2010sAA
group = 2010snonAA_US
group = 2010snonAA_nonUS
aae_feature
ing ultimas
ai monophthongization
post-vocalic r
post-consonantal d
post-consonantal t
auxiliary verb
third person singular
zero copula
0.00
0.25
0.50
0.75
1.00
aae_realization
song_type
cover
original